

Everest Engineering College

Sanepa, Lalitpur

Assignment 4 & 5 , 2026

Level : BE Software, 3rd Semester

Subject: Calculus II

Due Date: Feb 11, 2026

Students are required to give their answers in their own words as far as practicable and copy from others is strictly Prohibited.

Attempt all the Questions:

1. Find the fundamental period of the functions $\cos 2x$, $\sin \pi x$
2. Test whether following functions are odd or even. i. $\sin x + \cos x$, $2 - 3x^4 + \sin^2 x$, ii.

$$f(x) = \begin{cases} e^x & \text{if } 0 < x < 1 \\ e^{-x} & \text{if } -1 < x < 0 \end{cases}.$$

3. Find the Fourier series of $f(x) = |x|$ for $-\pi < x < \pi$ and then show that

$$\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}$$

4. Find the Fourier series of the function

$$f(x) = \begin{cases} k & \text{if } -\pi/2 < x < \pi/2 \\ 0 & \text{if } \pi/2 < x < 3\pi/2 \end{cases}.$$

having periodicity 2π . Also show that $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots = \frac{\pi}{4}$.

5. Find the Fourier series of $f(x) = |\cos x|$ for $-\pi < x < \pi$.
6. Find the Fourier series representation of the periodic function $f(x) = \frac{x^2}{2}$ for $-\pi < x < \pi$ and then show that
 - i. $1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \dots = \frac{\pi^2}{6}$
 - ii. $1 - \frac{1}{4} + \frac{1}{9} - \frac{1}{16} + \dots = \frac{\pi^2}{12}$
7. Find the Fourier series of the function

$$f(x) = \begin{cases} -x & \text{if } -2 < x < 0 \\ x & \text{if } 0 < x < 2 \end{cases}.$$

8. Find the Fourier series of

$$f(x) = \begin{cases} 0 & \text{if } -2 < x < -1 \\ k & \text{if } -1 < x < 1 \\ 0 & \text{if } 1 < x < 2 \end{cases}.$$

9. Find the Fourier series of the function $f(x) = x^2$ in the interval $-L < x < L$.
10. Find the Fourier series of $f(x) = \frac{x}{2}$ over the interval $0 < x < 2\pi$. Also,
 - i. Sketch the graph of $f(x)$ in the interval $0 < x < 4\pi$
 - ii. By taking an appropriate value of x , show that $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$
11. Find the Fourier series of the function

$$f(x) = \begin{cases} x & \text{if } 0 < x < \pi \\ \pi & \text{if } \pi < x < 2\pi \end{cases}.$$

Then

- i. Sketch a graph of $f(x)$ in the interval $-2\pi < x < 2\pi$.
 - ii. Show that $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$
 - iii. $\frac{\pi^2}{8} = 1 + \frac{1}{3^2} + \frac{1}{5^2} + \dots$
12. Find the Fourier Sine and Cosine series representation of the half range function $f(x) = x^2$ for $0 < x < L$.
 13. Find the Fourier Sine and Cosine series of $f(x) = x$ for $0 < x < \pi$.
 14. Find the Fourier Cosine series representation of the half range function $f(x) = x^2$ for $0 < x < \pi$ and deduce that $\frac{\pi^2}{6} = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots$

"ALL THE BEST."